① Screen 1 : Start Screen.

start game button : when clicked, navigate to "Player Setup Screen"

↳ where users enter Player names to begin the gam

Instruction button : when clicked, open a new screen showing game rules for al of

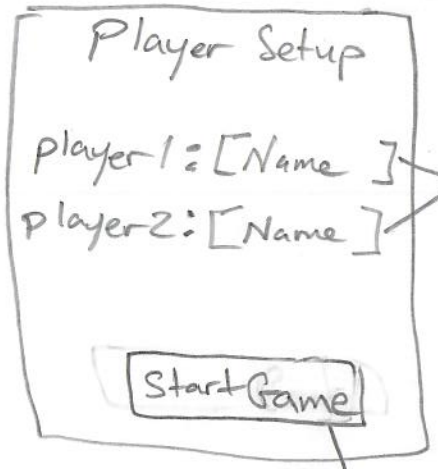
↳ place ships

↳ take turns

↳ hits / miss system

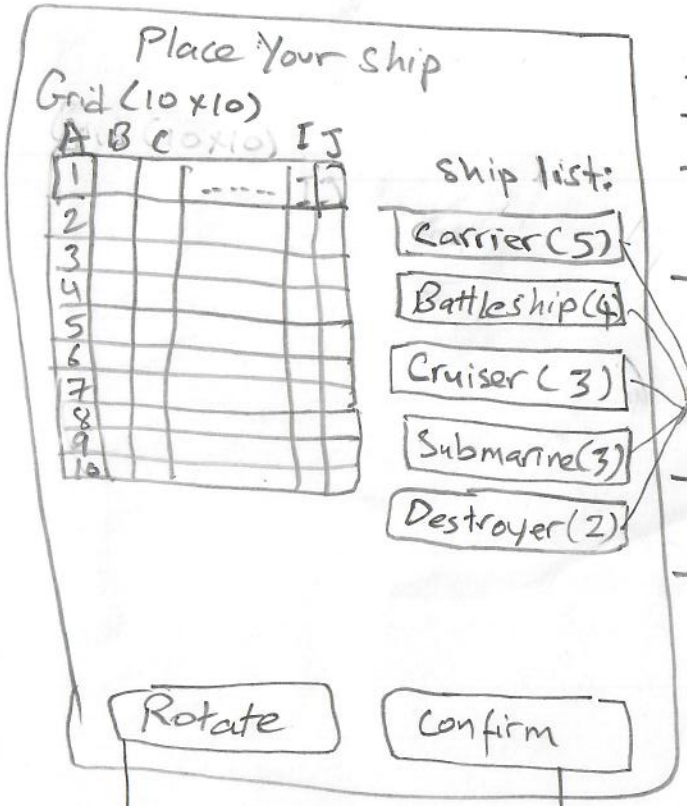
↳ win condition

② Screen 2: Player Setup Screen



- Players enters names before starting the game
  - default name can be assigned if left empty.
  - first input is assigned as "Player 1"
  - Second input is assigned as "player 2"
    - Player 1: start the game as default.
    - if NO name input → system randomize first player.
- start game button: moves to "ship Placement Screen".

Screen 3: "Ship Placement Screen"



→ players select a ship from the list before placing it on the grid.
→ player click grid cell to place ships.
→ each ship occupies multiple cells based on size
↳ Number represent ship size (number of grid cell)
→ Ships can be placed anywhere on the grid but within boundaries
↳ ships can be placed horizontally or vertically
↳ ships can not overlap
→ Each player must place exactly 5 ships before starting the game.
→ Each ship can be placed only once.
→ selected ship is highlighted.
→ placed ships are marked as completed (✓).
⟹ when all ships are placed, we can click "confirm" button
↳ confirm button is enabled only after all ships are placed.
↳ When clicked: ship positions are locked and can not be changed.

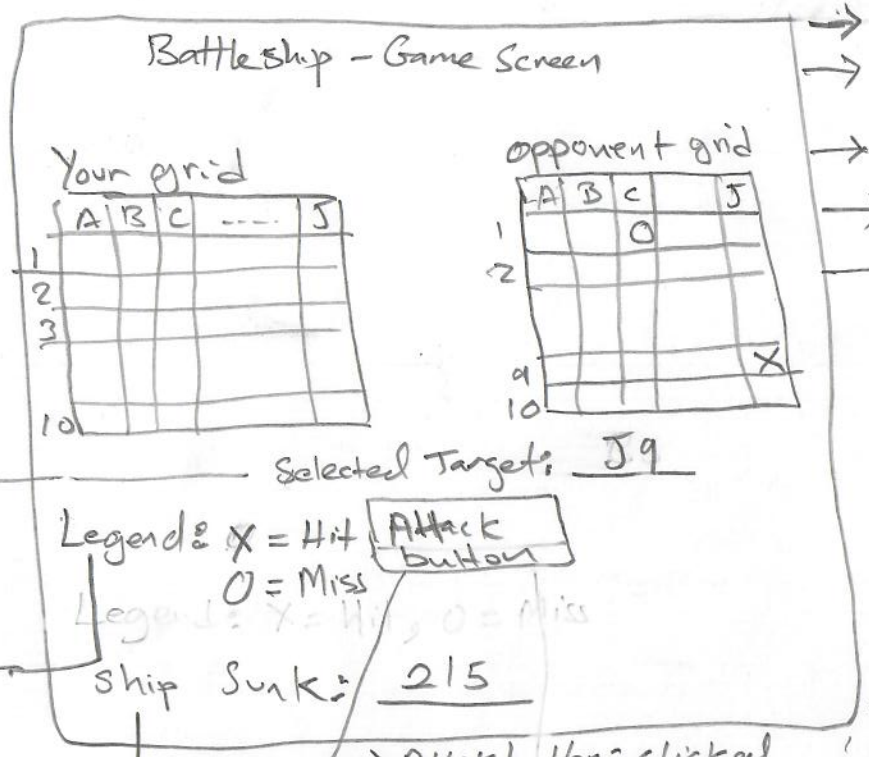
"button"

↳ Rotate button:
↳ change ship direction (horizontally/vertically)

* Additional features:
① System prevent selecting already placed ships.
② System highlight vaild placement position on grid.
③ System show message "Invalid placement, ship overlap, or out of boundaries" if placement is not correct.

Screen 4: Game play screen

↳ this screen is used to: show whose turn it is, let the player attack, show hits & miss, track progres



→ the screen display both players own grid and the opponent grid.

→ the player select one coordinates on the opponent grid during their turn.

→ the selected coordinates is shown before the attack is submit

→ the attack button submit the selected coordinates.

→ the system display X for hit and O for miss.

→ System compare selected coordinate with Opponent sh position.

↳ if coordinate contain a ship → system marks X (hi

↳ if coordinate does Not contain a ship

↳ system marks O (miss)

→ System store attack result for future reference.

↳ each coordinate can be only attacked once.

→ The opponent's ship are hidden unless hit.

↳ Ships are revealed only when hit occurs.

→ the system prevent the player from attacking the same coordinates twice → message is displayed.

→ after one attack, the turn switches to the other player.

→ the screen shows how many ships have been sunk.

↳ if ship is fully hit, system mark it as sunk

→ the game continue until one player sink all 5 ships.

→ Attack button: clicked after selecting a target on the opponent grid

→ shows progress in the game → track destroyed ship

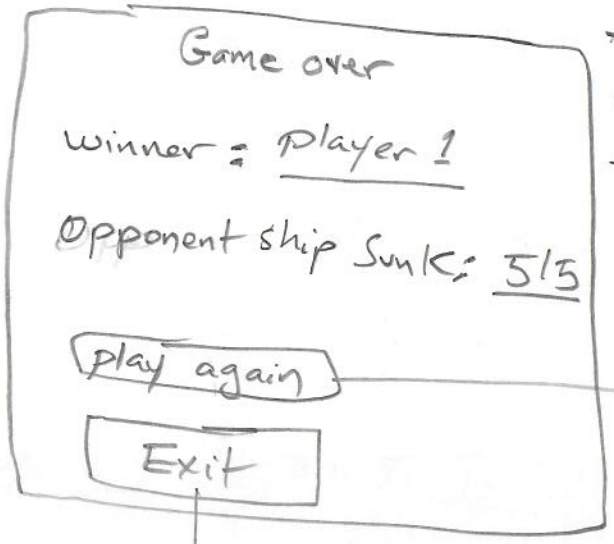
→ shows which coordinates the player chose to target opponent before attacking.

→ statis display, used for clarity and usability.

Screen 5: Game Result Screen

↳ this screen appear when: One player has sunk all 5 opponent ships.

↳ system announce the winner.



→ screen is displayed when one player sinks all five opponent ships.

→ the system announce the winner

→ No more moves can be made after this screen appears.

→ play Again start a new match (game)

→ Exit closes the game or returns to the menue.

→ play again button: return users to start Screen or play setup page.

↳ Exit button: close the game.

* system can display a short summary of game performance.

Like:

	player 1	player 2
Total hits =	_	_
Total Miss =	_	_
Ship Sunk =	_	_